

Kandice Lu

☎ 413-834-8510 ✉ kandicelu3.14@gmail.com Portfolio: <https://kandicelu.github.io/>

EDUCATION

Cornell University – B.S. in Mechanical Engineering

May 2026

- GPA: 4.11/4.3, 3.93/4.0, Awards: *Sibley Prize*

Cornell University – M.Eng in Aerospace Engineering

Expected Dec 2026

RELEVANT COURSEWORK

- Finite Element Analysis, Vibrations, Material Mechanics, Composites, Fluid Mechanics, Thermodynamics, Heat Transfer, Engineering Computation

SKILLS

- **Technical Skills:** CAD, FEA, Structural Design and Analysis, Data Analysis, Machining, Welding, DFMA, GD&T
- **Software:** ANSYS, Solidworks, Fusion 360, STAAD.Pro, LabVIEW
- **Programming Languages:** MATLAB, C/C++, Python

ENGINEERING EXPERIENCE

Full Team Lead, Fabrication Lead

Sep 2023 – Present

Cornell Steel Bridge Project Team

Ithaca, NY

- Led a multi-disciplinary team of 30 students to design and fabricate a 23 foot steel bridge to withstand a 2500lb load with minimal deflection, placing third in a regional competition
- Analyzed design variations with FEA to optimize geometries and section properties, improving projected structural efficiency costs by 15%
- Designed and validated connection geometries using CAD and FEA, ensuring structural integrity under worst-case loads
- Performed final assessment of all designed components to ensure compatibility with adjoining parts, in-house manufacturability, and compliance with design requirements
- Assessed structural performance of final assembly with static load testing, implementing necessary fixes based on observations to reduce sway by 50%
- Managed overall project logistics, handling material procurement, coordinating work across sub-teams, and ensuring adherence to established timelines

Undergraduate Researcher

May 2025 – Dec 2025

ZT Group

Ithaca, NY

- Developed and optimized fabrication processes for polymer-based thermal interface materials
- Characterized thermal properties with laser flash testing to guide iterative sample creation

COURSEWORK PROJECTS

Finite Element Analysis: Wing Design

Oct 2025 – Dec 2025

- Designed an airplane wing to minimize weight while meeting deflection and safety factor requirements
- Analyzed and optimized various design configurations in ANSYS, reducing initial weight by over 20%
- Performed manual and automatic mesh refinement tests to ensure convergence and verification of results

Fluids Lab: Wind Turbine Blade Design

Oct 2025 – Dec 2025

- Designed and tested a small scale wind turbine
- Constructed a mathematical model of blade performance in MATLAB
- Validated model and physical turbine performance through wind tunnel testing

Introduction to Aeronautics: Glider Project

Sep 2024 – Dec 2024

- Designed a slow flying balsa glider, performing lift and drag calculations to achieve balanced flight
- Modeled glider skeleton with CAD and constructed glider via laser cutting and hand assembly
- Tested and rebalanced constructed glider to achieve desired speed and stability, achieving 5th place in final competition

WORK EXPERIENCE

Finger Lakes Reuse

June 2025 – Aug 2025

Administrative Project Intern

Ithaca, NY

- Researched the viability of and assisted in the implementation of transitions to new report creation and price management systems
- Assisted chief officers in creating and organizing various strategic documents, databases, and reports